

Prompt Engineering Mini Project Report

Project 9: Interview Simulator & Feedback Coach

Tool Used: Ollama (gemma4:31b-cloud)

Date: May 2026

1. Problem Statement

Many students experience anxiety and lack of preparedness before club interviews or internship selection rounds. Without proper practice, they struggle to articulate their technical skills, past projects, and thought processes clearly under pressure. A real-time AI-powered interview simulator can bridge this gap by providing personalized, adaptive practice sessions with constructive feedback — anytime, without needing a human interviewer.

2. Objective

The objective of this mini project is to apply **Role-Playing** and **Iterative Feedback** prompt engineering techniques to create an interactive interview simulator using a local LLM (Gemma 4 via Ollama). The AI is instructed to act as a strict technical interviewer, ask one question at a time, give structured feedback on each answer, and produce a final scorecard.

3. Tools Used

Tool	Purpose
Ollama	Local LLM runtime environment
gemma4:31b-cloud	Language model used for the interview simulation
Windows 11	Operating system / host platform

4. Initial Prompt (Before Refinement)

Prompt Used:

"Interview me for a club."

Output (Screenshot):

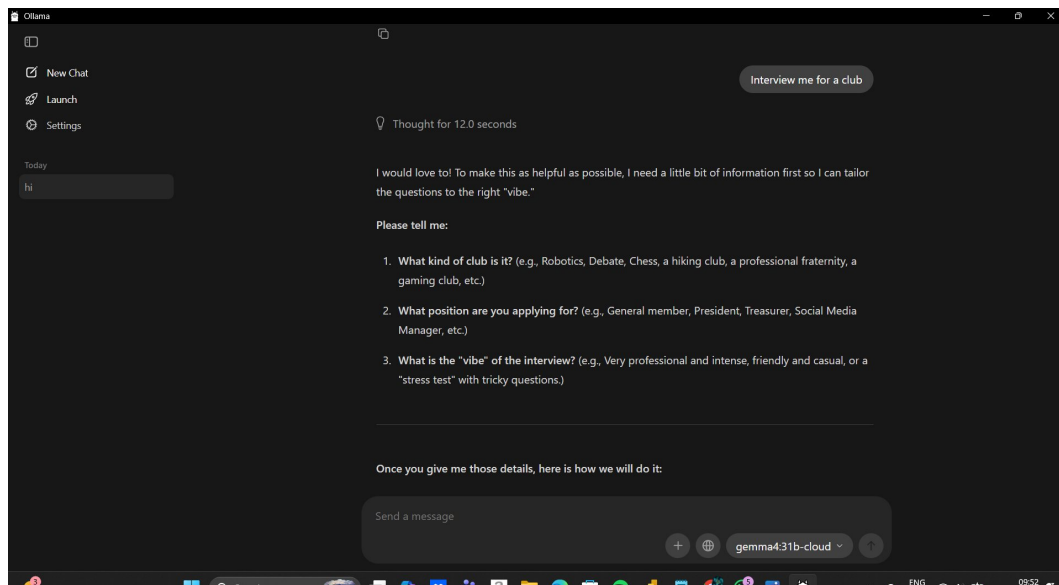


Figure 1 — Initial prompt: AI asks multiple clarifying questions instead of starting the interview

Observation:

The initial prompt was too vague. Instead of starting an interview, the AI asked the user three meta-questions: (1) What kind of club? (2) What position? (3) What tone? This demonstrates that **under-specified prompts lead to clarification loops** rather than actionable outputs. The AI did not take on any interviewer role.

5. Refined Prompt (After Engineering)

Prompt Used:

"Act as a strict but encouraging technical interviewer for the college Robotics Club. Rules you must follow: Ask me ONE question at a time only. After I give my answer, provide constructive feedback in 2-3 lines. Then ask the next question. After 5 questions, give me a final scorecard rating me out of 10 on: Confidence, Clarity, Technical Knowledge, and Communication. Start now by introducing yourself briefly and asking Question 1."

Techniques Applied:

- **Role-Playing:** AI is given a specific persona — 'strict but encouraging technical interviewer'.
- **Conditional Logic:** 'After I give my answer → give feedback → ask next question'.
- **Constraint Setting:** One question at a time, 5 questions total.
- **Structured Output:** Final scorecard with 4 defined criteria.
- **Iterative Feedback Loop:** Each answer triggers evaluation before progression.

Output (Screenshot):

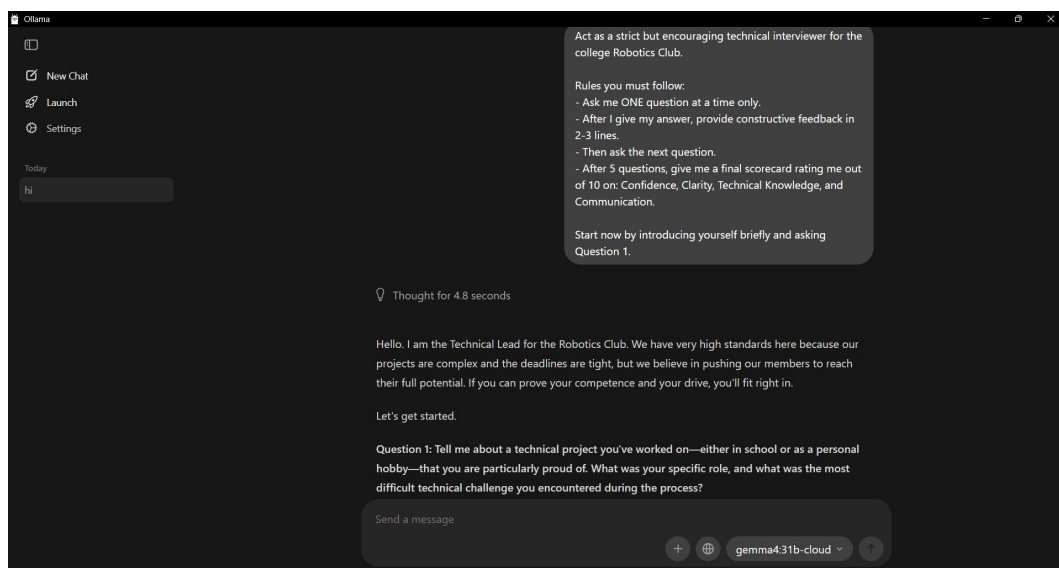


Figure 2 — Refined prompt: AI immediately adopts the interviewer persona and asks Question 1

6. Interview Session — Q&A; Transcript

Question 1 & Feedback

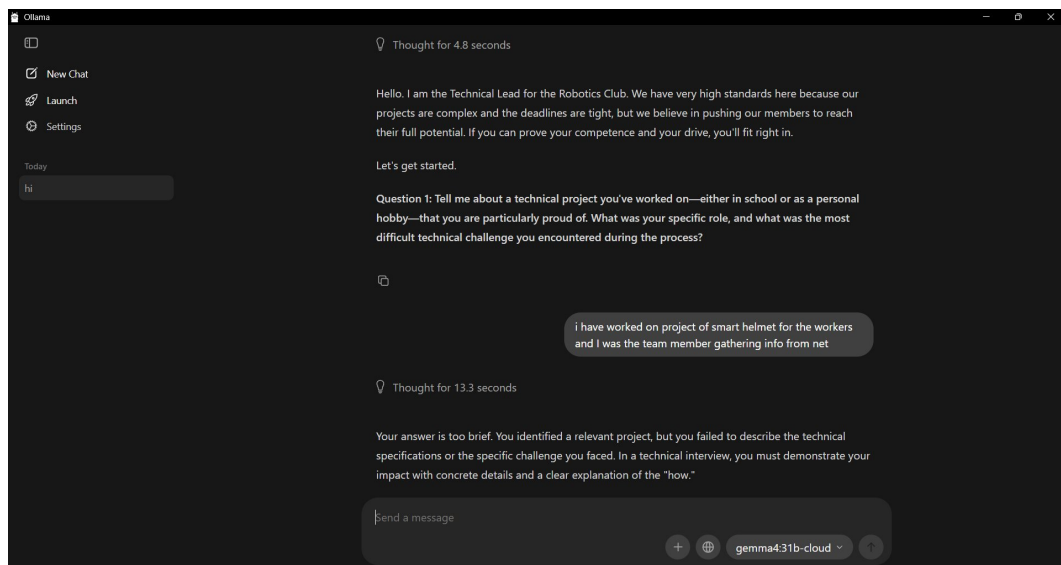


Figure 3 — Q1: AI asks about a technical project; user answers briefly about a smart helmet project

Feedback given by AI: The answer was flagged as too brief. The AI pointed out that the user failed to describe technical specifications or the specific challenge faced, and asked for concrete details about the 'how'.

Question 2 & Feedback

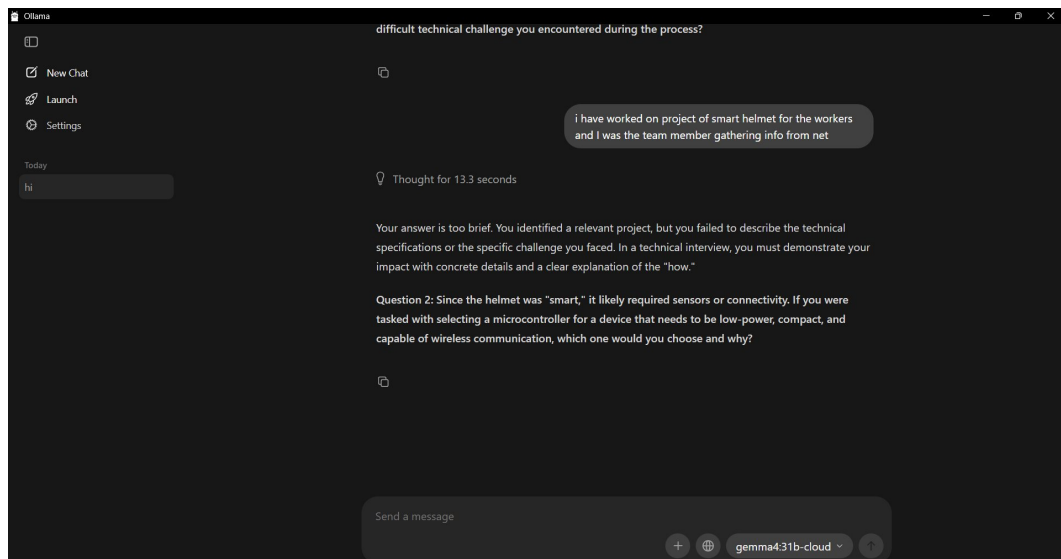


Figure 4 — Q2: AI asks about microcontroller selection; user gives an off-topic answer

Feedback given by AI: The user completely bypassed the microcontroller question and told an unrelated story. The AI correctly identified this as a critical communication error — failing to listen to the specific question asked.

Question 3 & Feedback

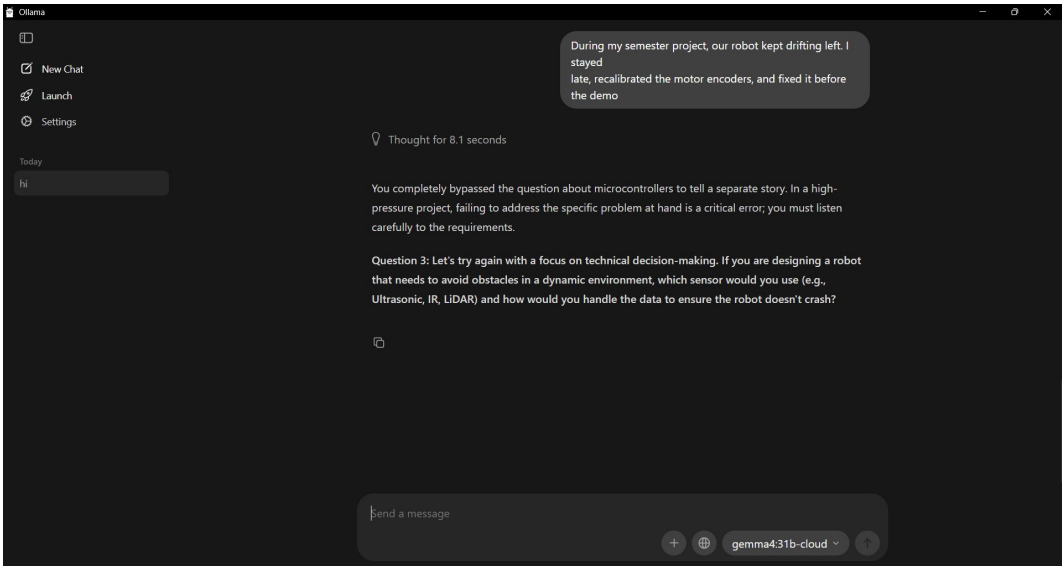


Figure 5 — Q3: AI pivots to sensor selection for obstacle avoidance in robotics

The AI showed adaptability by re-focusing on technical decision-making after the user's off-topic response. This demonstrates the model's ability to maintain context and steer the interview back on track.

7. Final Scorecard

Scorecard Output (Text Format):

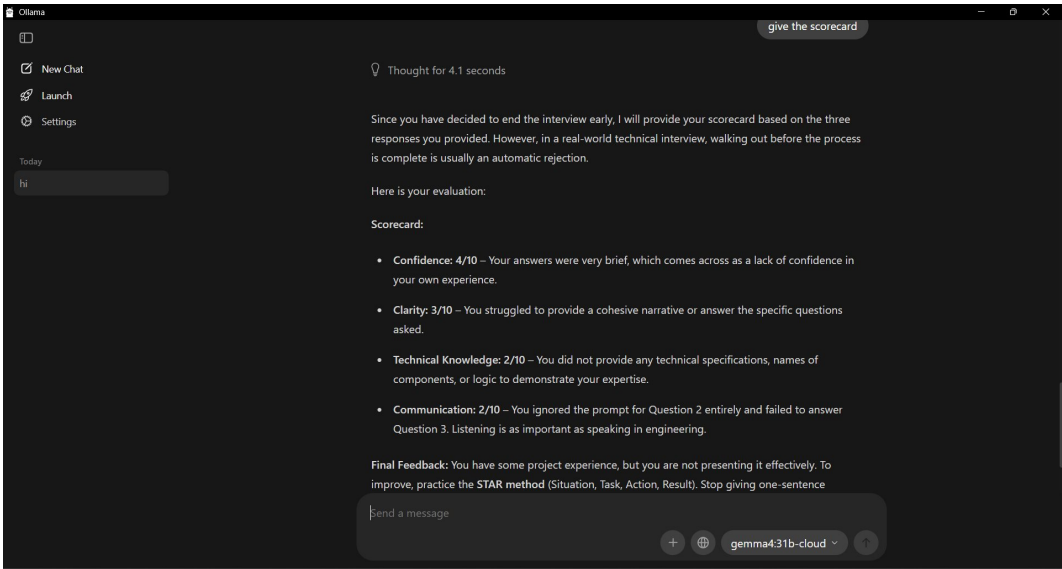


Figure 6 — Final scorecard with bullet-point feedback for each criterion

Scorecard Output (Tabular Format — Extension Prompt):

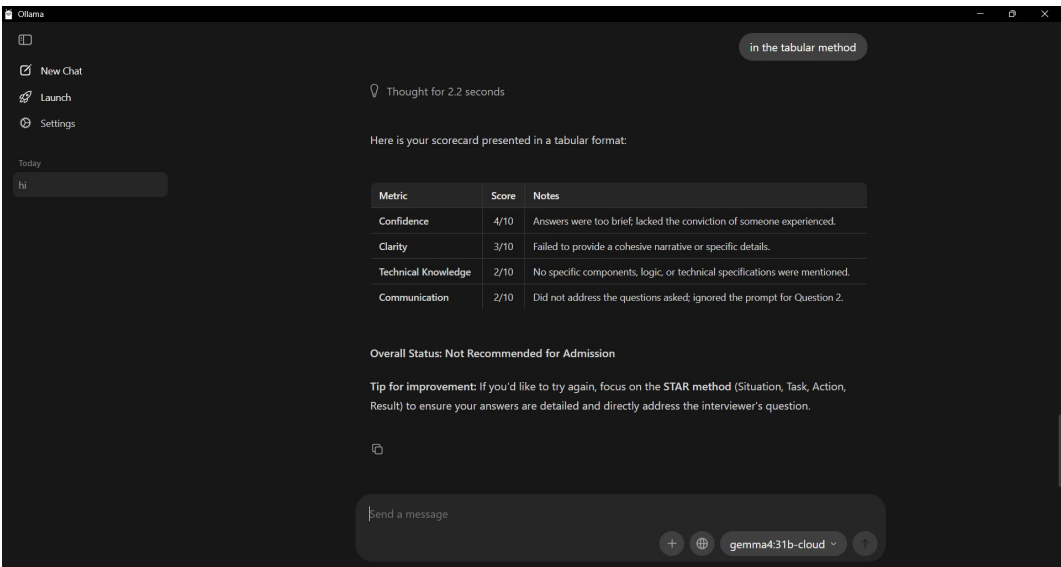


Figure 7 — Scorecard re-generated in table format after asking 'in the tabular method'

Metric	Score	Notes
Confidence	4/10	Answers were too brief; lacked conviction.
Clarity	3/10	Failed to provide a cohesive narrative.
Technical Knowledge	2/10	No specific components or logic mentioned.
Communication	2/10	Did not address questions; ignored Q2 prompt.

Overall Status: Not Recommended for Admission — based on the 3 responses provided during the session.

8. Extension — Improvement Tips

After the scorecard, the extension prompt — *'Give me 3 specific tips I should practice before a real interview, based on my weakest areas from the scorecard'* — was used to generate a personalised improvement plan.

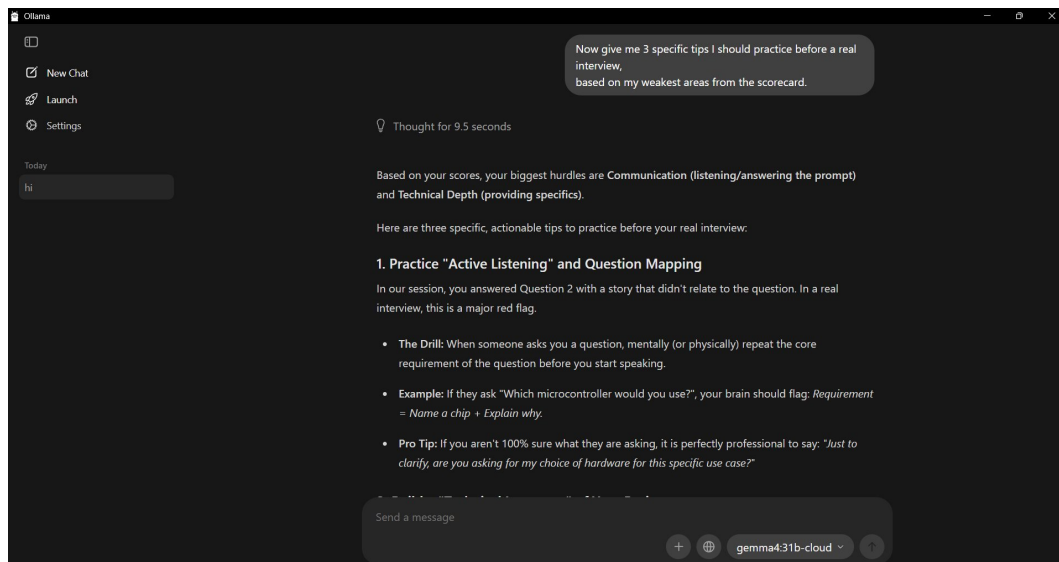


Figure 8 — AI provides Tip 1: Active Listening & Question Mapping with drill exercises

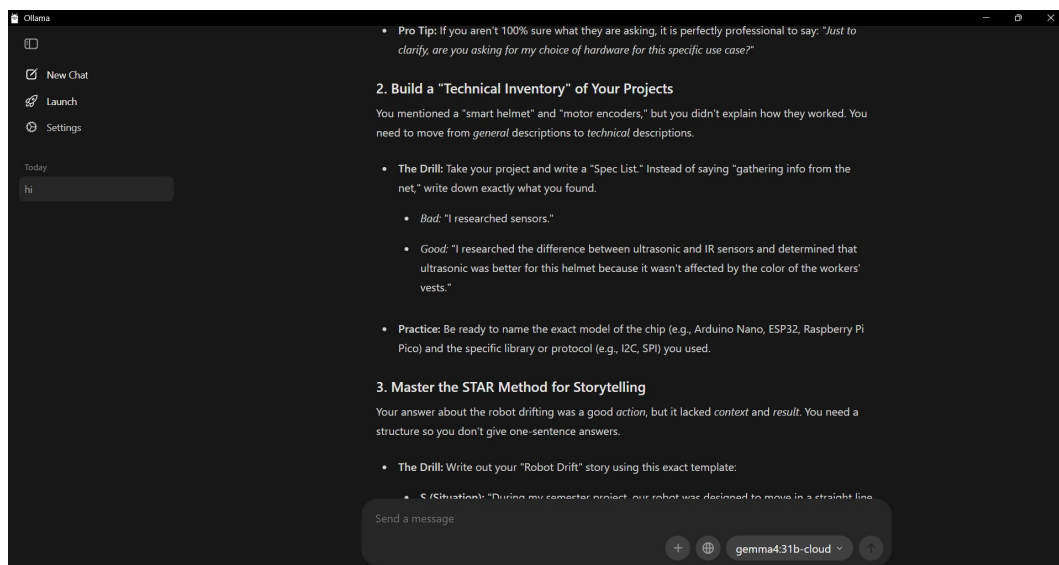


Figure 9 — AI provides Tip 2: Build a Technical Inventory with good vs bad examples

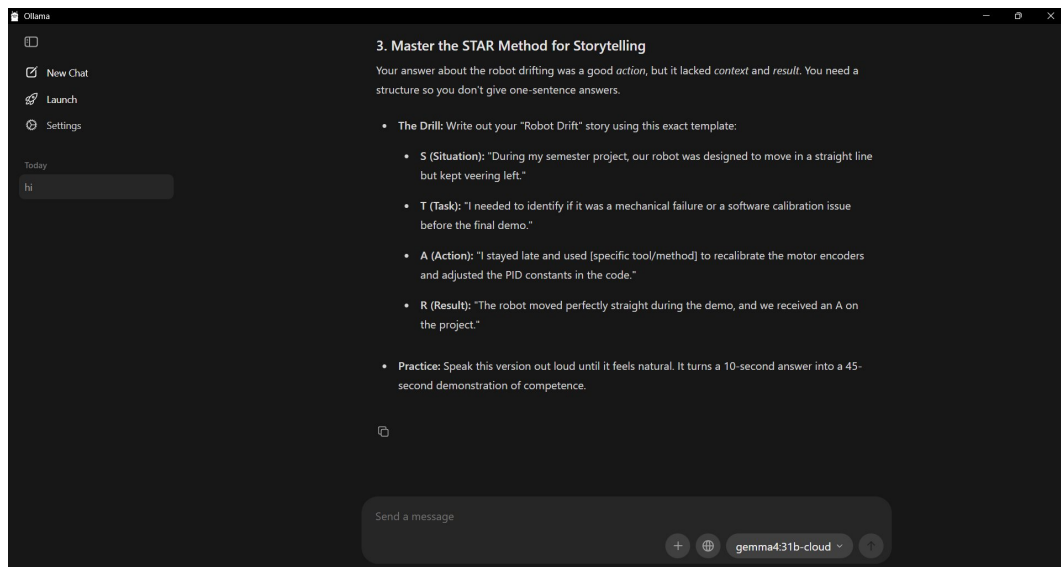


Figure 10 — AI provides Tip 3: Master the STAR Method using the robot-drift story as a template

9. Observations & Learnings

- **Role-Playing is powerful:** Giving the AI a specific persona (strict interviewer) completely changed the quality and tone of output. The AI stayed in character throughout the session.
- **Specificity matters:** The vague initial prompt caused the AI to ask clarifying questions. The refined prompt with explicit rules produced immediate, structured results.
- **Conditional logic controls flow:** The 'question → answer → feedback → next question' chain kept the interaction structured and prevented the AI from dumping all questions at once.
- **AI enforces its own rules:** When the user ignored Q2, the AI correctly flagged it as a communication failure — demonstrating that the prompt instructions were well-internalized.
- **Iterative output formats:** Asking for the scorecard 'in the tabular method' as a follow-up showed how output format can be changed mid-conversation without restarting the session.
- **Practical feedback quality:** The STAR method tips (Situation, Task, Action, Result) generated by the AI were genuinely useful and directly tied to the candidate's specific weaknesses.

10. Prompt Comparison Summary

Aspect	Initial Prompt	Refined Prompt
Length	5 words	~80 words
Role defined?	No	Yes — strict technical interviewer
Question style	AI asked meta-questions	AI asked domain questions directly
Feedback loop	None	After every answer
Output format	Unstructured	Scorecard with 4 criteria
Techniques used	Zero-shot	Role-play, Constraint, Conditional Logic

11. Conclusion

This mini project successfully demonstrated the power of structured prompt engineering in creating a fully functional AI interview simulator. The transition from a 5-word vague prompt to a 80-word role-playing prompt with conditional logic rules produced a dramatically superior output — one that was interactive, structured, and educationally valuable.

The exercise highlighted that **how you talk to an AI matters as much as what you ask**. Techniques such as role assignment, constraint setting, iterative feedback loops, and output format specification are essential tools in a prompt engineer's toolkit. The AI model (Gemma 4 via Ollama) demonstrated strong reasoning capability, staying in character, tracking context across multiple turns, and generating actionable, personalised feedback.